NEERAJ MUDUNURU

Cybersecurity Researcher | Digital Forensics, SOC & AI-Driven Security Solutions

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PROFESSIONAL SUMMARY

Cybersecurity researcher (B.Tech Cyber Security, 2027) focused on digital forensics, incident response and AI-driven security tooling. Published a first-hand investigation of a LockBit-branded ransomware intrusion, supported live cybercrime cases at the AP Police Cyber Crime Department, and monitored client endpoints with Wazuh in a SOC internship. Builds and publishes open-source security tools (CI-integrated DAST, leaked-secret blast-radius analysis, ML intrusion detection) and LLM applications with human-in-the-loop review, and researches SIEM integrations for Wazuh.

TECHNICAL SKILLS

Digital Forensics: KAPE, EvtxECmd, MFTECmd, AmcacheParser, RegRipper, Plaso (log2timeline), Chainsaw (Sigma), Hindsight, FTK Imager, FTK Toolkit, Hancom Forensic Tools, Oxygen Mobile Forensics, CDR/IPDR Analysis, Windows Artifact Analysis (EVTX, Amcache, Registry, MFT, UserAssist), Timeline Analysis, Chain of Custody, LSB Steganography

Security Operations: Wazuh (custom decoders, rules, CDB lists), Splunk (search, reporting), SIEM Integrations, Threat Detection, Incident Response, Log Analysis, Network Traffic Monitoring, VirusTotal, Shodan, urlscan.io

AI & Automation: Python, FastAPI, LangChain (LCEL), LangGraph, LangSmith, Generative AI, RAG, ChromaDB, SentenceTransformers, NLP, Streamlit, scikit-learn, AI-Assisted SOC Automation

Application & Pipeline Security: DAST, OWASP ZAP, SAST, Vulnerability Scanning, Reconnaissance, SARIF, GitHub Actions, GitLab CI, Azure DevOps, Docker

Other: Malware Analysis Fundamentals, SCADA/ICS Security, AI-Based Attack and Defense, TCP/IP, WireGuard VPN, DPDP Act 2023

PROFESSIONAL EXPERIENCE

Cyber Security Volunteer | Cyber Secured India *September 2026 – Present*

- Coordinate speaker engagement and organize cybersecurity programs, supporting the team in delivering successful sessions.
- Guide and mentor a cohort of 100 interns learning cybersecurity.
- Teach digital forensics fundamentals to interns and support their practical learning.

Cyber Law Technical Consultant | SIRI Law LLP *June 2026 – September 2026*

- Worked in the firm's Cyber and Digital Law practice, reporting to the Managing Partner; received a Letter of Recommendation.
- Assessed technology files against the Digital Personal Data Protection (DPDP) Act 2023, identifying what personal data was processed, by whom, and the resulting legal questions; prepared working notes used by the legal team.
- Delivered technical documentation and auditing on two client engagements, flagging gaps and inconsistencies for the legal team.
- Performed forensic analysis on one engagement and guided legal team members so written findings matched the technical evidence.

SOC Analyst Intern | WeAreCyberSec *February 2026 – Present*

- Monitored and triaged security events across 10 client endpoints using Wazuh; maintained the Wazuh infrastructure and resolved endpoint connectivity issues.
- Designed and deployed a WireGuard-based monitoring architecture that routed Wazuh telemetry through an encrypted VPN without disrupting normal endpoint traffic; integrated VirusTotal threat intelligence and automated reporting with Python and Discord webhooks.
- Conducted a LockBit-branded ransomware forensic investigation across 1 Windows Server and 3 Windows endpoints using KAPE, analyzing EVTX, Amcache, Registry, MFT, AnyDesk and SMB artifacts; identified a Windows endpoint as patient zero and reconstructed the path to the Windows Server where the ransomware executed.
- Responded to a DDoS attack on a client environment: analyzed abnormal traffic, identified malicious sources, implemented defensive controls and coordinated mitigation to restore service.
- Developing AI-assisted SOC automation for alert triage, investigation support and report generation; contributed authorized security testing and code-level recommendations to an AI-powered penetration-testing product.

Agentic AI Intern | Innomatics Research Labs *February 2026 – April 2026*

- Built an AI resume-screening system with LangChain pipelines for skill extraction, matching and 0–100 scoring with explanations; traced every run with LangSmith.
- Developed a RAG customer-support assistant with LangGraph, ChromaDB and SentenceTransformers, including intent classification and human-in-the-loop escalation.
- Built an end-to-end AI chatbot pipeline using NLP techniques including sentiment analysis and text normalization.
- Delivered a FastAPI e-commerce backend with REST API integration; solved 100+ HackerRank challenges.

Cyber Security Engineer | COE Security LLC, North Carolina, USA (Night Shift) *December 2025 – May 2026*

- Managed end-to-end onboarding for 5+ security engagements, coordinating requirements between clients and internal teams to reduce onboarding delays.
- Used Splunk for search, report generation and routine operations tasks.
- Conducted OSINT research on key stakeholders and certification bodies, delivering intelligence reports that supported business development and pre-engagement planning.
- Maintained project documentation covering client requirements, workflow updates and progress logs.
- Enhanced an AI-based penetration testing tool by researching remote IoT attack surfaces, recommending security hardening and configuring secure VPN tunnels for controlled client-side testing.

Internship Mentor | Cyber Security & Digital Forensics UK Ltd *October 2025 – January 2026*

- Led and mentored a five-member cyber security team from planning and task allocation through milestone delivery.
- Delivered 10+ executive-level presentations on project progress, threat findings and security outcomes to senior stakeholders.
- Coached team members on security tools, methodologies and problem-solving frameworks.

Cyber Crime Investigative Intern | Cyber Crime Department, AP Police *July 2025 – August 2025*

- Performed CDR and IPDR analysis on live cybercrime cases, mapping communication timelines and identifying suspicious patterns that guided active investigations.
- Used Hancom Forensic Tools and Oxygen Mobile Forensics to extract, preserve and analyze mobile device evidence while maintaining chain of custody.
- Investigated a fake online betting application fraud case, tracing digital footprints and operational patterns; findings contributed to case resolution.
- Worked with law enforcement analysts to produce structured forensic reports for judicial proceedings.

Cyber Security Intern | Cyber Secured India *April 2025 – June 2025*

- Conducted reconnaissance and vulnerability scanning on real-world cybercrime case studies and produced threat analysis reports.
- Used FTK Imager and FTK Toolkit for evidence acquisition, media imaging and artifact analysis with chain-of-custody documentation.
- Scored 92.81% and ranked in the Top 50 among 5,000+ participants in a hands-on security assessment, exploiting CVE-2022-26965 in Pluck CMS 4.7.16 for authenticated remote code execution and privilege escalation to root.

RESEARCH & PROJECTS

Security Workbench: Public Security Research Repository | github.com/Rolex-1905/security-workbench

- LockBit-Branded Ransomware Investigation (write-up)
  - Triaged a 793-file KAPE collection using Chainsaw, EvtxECmd, MFTECmd, AmcacheParser, RegRipper and Plaso to build a unified attack timeline.
  - Wrote a custom Python parser for AnyDesk's UTF-16 connection log, surfacing a remote device that connected exactly once, at the start of the attack window, against 77 sessions for the legitimate admin device.
  - Corroborated with UserAssist, SAM and MFT artifacts; found antivirus protections disabled just before the payload ran and 92 of 158 event logs empty, pointing to deliberate log tampering.
  - Quantified impact (9,138 encrypted files, 1,678 ransom notes) and hypothesized an independent actor using a leaked LockBit 3.0 builder.
- Wazuh SIEM Integrations Reference (write-up)
  - Researched eight external integrations with Wazuh (Have I Been Squatted, Filescan.io, Shodan, urlscan.io, InsecureWeb, Symantec EDR, CrowdStrike Falcon, Sophos Firewall) from published community posts that did not work as written, and implemented a subset.

  - Corrected defects in the source material, including a decoder regex error that would have broken most Sophos field extraction, duplicate rule IDs in the Symantec ruleset, a flattened CrowdStrike severity ladder and hardcoded API keys; renumbered all rules into collision-free ID blocks and added testing steps.

Aysal-Scan: Secret Detection and Blast Radius Analysis | github.com/Rolex-1905/aysal-scan

- Published to PyPI; detects 17+ secret types across files, git history and GitHub URLs using regex and Shannon entropy.
- Built a blast-radius engine that verifies whether leaked keys are still active and enumerates their permissions through read-only provider API calls (AWS, GitHub, OpenAI, Stripe, Slack and others).
- Added severity scoring, remediation guidance, SARIF output, a GitHub Action and a pre-commit hook.

Aysal-Shield: DAST Automation Platform for Apache Tomcat | github.com/Rolex-1905/aysal-shield

- Built a CLI-first platform that orchestrates OWASP ZAP with authenticated crawling, Tomcat hardening checks aligned with CIS benchmarks, finding normalization and deduplication, and JSON/HTML/CSV/SARIF reporting.
- Integrated policy-based CI gates for GitHub Actions, GitLab CI and Azure DevOps that fail builds on High/Critical findings; added authorization enforcement, PII redaction and Docker packaging.

Aysal-IDS: ML Network Intrusion Detection | github.com/Rolex-1905/Aysal-IDS

- Trained Random Forest and Logistic Regression classifiers on CICIDS2017 (about 2.8M flows), reaching 99.93% accuracy.
- Validated against CICIDS2018 and documented the generalization gap (77.08% accuracy, 3% attack recall), which shaped a V2 roadmap for adaptive retraining.

RAG-Based Customer Support Assistant with HITL | github.com/Rolex-1905/rag-customer-support-assistant

- Built a retrieval-augmented system with a 7-node LangGraph workflow, SentenceTransformers embeddings and a ChromaDB vector store over PDF knowledge bases.
- Implemented intent classification and confidence-based routing that escalates to a human agent when retrieval confidence falls below 35%, the user asks for a human, or no relevant context is found; includes HLD/LLD documentation and a test suite.

AI Resume Screening System (LangChain, LangSmith) | github.com/Rolex-1905/ai-resume-screening-system

- Built an LCEL pipeline (extract, match, score) that converts resumes into structured data, scores candidates 0–100 against a job description, and generates explanations.
- Added LangSmith tracing per resume, a Streamlit comparison dashboard with candidate ranking and skill-gap analysis, and PDF/TXT parsing.

EDUCATION

Bachelor of Technology in Cyber Security *2023 – 2027*

Vignan's Institute of Information Technology, Visakhapatnam | CGPA: 8.80/10

CERTIFICATIONS

- Greyhawk Certified Digital Forensics Practitioner, Greyhawk Academy (Oct 2026, 40-hour course with scenario-based exam) | Verify
- Cybersecurity Defense Analyst Career Path, Cisco Networking Academy with Splunk (Oct 2026) | Credly badge
- Mastering GenAI Tools for SOC Analysts, SOCRadar (April 2026) | Credsverse certificate
- CCNA: Introduction to Networks, Cisco Networking Academy | Credly badge
- Introduction to Networking for Cyber Professionals, Zscaler Academy (March 2025) | Skilljar certificate
- OPSWAT Academy: Critical Infrastructure Protection (ICIP), Network Security Associate (ONSA), File Security Associate (OFSA), Web Traffic Protection Associate (OWPA), Email Security Associate (OESA), Secure Storage Associate (OSSA), Legacy Systems Security Associate (OLSA)

COMPETITIONS & OPEN SOURCE

- Contributor, Open Source Connect India 2026 and Open Source Connect Global 2026

LANGUAGES

English (Intermediate, B2) | Telugu (Advanced, C1)